GENERAL SCIENCE: PHYSICS FUNDAMENTALS

QUANTITY UNIT AND VECTOR → MECHANICS CAUSE-AND-RATE RAIL → ORBIT AND FREE-FALL MAP → FLUID-PRINCIPLE SELECTION MATRIX → THERMAL STATE AND TRANSFER → WAVE SOUND AND SPECTRUM

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 QUANTITY UNIT AND VECTOR DECISION TREE

QUANTITY UNIT AND VECTOR DECISION TREE | SCALAR OR VECTOR? | BASE OR DERIVED SI? | CONSISTENCY CHECK ONLY | CALIBRATION + ORIENTATION + UNCERTAINTY | DISTANCE, NEVER TIME

QUANTITY → scalar or vector?

UNIT → base or derived SI? → DIMENSION → consistency check only → INSTRUMENT → calibration + orientation + uncertainty

CONCEPT / ARCHITECTURE
- QUANTITY → scalar or vector?
- UNIT → base or derived SI?

MECHANISM / APPLICATION
- DIMENSION → consistency check only
- INSTRUMENT → calibration + orientation + uncertainty

READINESS / STATUS LIMIT
- LIGHT-YEAR → distance, never time
- QUANTITY → scalar or vector?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INSTRUMENT → calibration + orientation + uncertainty; LIGHT-YEAR → distance, never time.

01

STAGE 01 MECHANICS CAUSE-AND-RATE RAIL

MECHANICS CAUSE-AND-RATE RAIL | NET FORCE | MOMENTUM CHANGE | FORCE X TIME | CHANGE IN MOMENTUM | FORCE + DISPLACEMENT | NET WORK | KINETIC-ENERGY CHANGE

NET FORCE → momentum change → IMPULSE → force x time → change in momentum → FORCE + DISPLACEMENT → work → NET WORK → kinetic-energy change → POWER → work or energy per time

CONCEPT / ARCHITECTURE
- NET FORCE → momentum change
- IMPULSE → force x time → change in momentum

MECHANISM / APPLICATION
- FORCE + DISPLACEMENT → work
- NET WORK → kinetic-energy change

READINESS / STATUS LIMIT
- POWER → work or energy per time
- NET FORCE → momentum change

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IMPULSE → force x time → change in momentum.

02

STAGE 02 ORBIT AND FREE-FALL MAP

ORBIT AND FREE-FALL MAP | CENTRIPETAL ACCELERATION | CONTINUOUS FREE FALL | EQUATORIAL + MATCHING ROTATION PERIOD | NEAR-POLAR LOW-ORBIT FAMILY | ESCAPE VELOCITY | CENTRAL BODY'S MASS AND RADIUS

GRAVITY → centripetal acceleration

ORBIT → continuous free fall → GEOSTATIONARY → equatorial + matching rotation period → POLAR / SUN-SYNCHRONOUS → near-polar low-orbit family

CONCEPT / ARCHITECTURE
- GRAVITY → centripetal acceleration
- ORBIT → continuous free fall

MECHANISM / APPLICATION
- GEOSTATIONARY → equatorial + matching rotation period
- POLAR / SUN-SYNCHRONOUS → near-polar low-orbit family

READINESS / STATUS LIMIT
- ESCAPE VELOCITY → central body's mass and radius
- GRAVITY → centripetal acceleration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ESCAPE VELOCITY → central body's mass and radius.

03

STAGE 03 FLUID-PRINCIPLE SELECTION MATRIX

FLUID-PRINCIPLE SELECTION MATRIX | CONFINED-FLUID PRESSURE TRANSMISSION | DISPLACED-FLUID WEIGHT AND UPTHRUST | PRESSURE-FLOW RELATION UNDER ASSUMPTIONS | PRESSURE COOKER | PRESSURE UP, BOILING TEMPERATURE UP | PRESSURE DOWN, BOILING TEMPERATURE DOWN

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PASCAL → confined-fluid pressure transmission	BERNOULLI → pressure-flow relation under assumptions	ALTITUDE → pressure down, boiling temperature down
ARCHIMEDES → displaced-fluid weight and upthrust	PRESSURE COOKER → pressure up, boiling temperature up	PASCAL → confined-fluid pressure transmission

CONCEPT / ARCHITECTURE
- GRAVITY → centripetal acceleration
- ORBIT → continuous free fall

MECHANISM / APPLICATION
- GEOSTATIONARY → equatorial + matching rotation period
- POLAR / SUN-SYNCHRONOUS → near-polar low-orbit family

READINESS / STATUS LIMIT
- ESCAPE VELOCITY → central body's mass and radius
- GRAVITY → centripetal acceleration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ESCAPE VELOCITY → central body's mass and radius.

03

STAGE 03 FLUID-PRINCIPLE SELECTION MATRIX

FLUID-PRINCIPLE SELECTION MATRIX · CONFINED-FLUID PRESSURE TRANSMISSION · DISPLACED-FLUID WEIGHT AND UPTHRUST · PRESSURE-FLOW RELATION UNDER ASSUMPTIONS · PRESSURE COOKER · PRESSURE UP, BOILING TEMPERATURE UP · PRESSURE DOWN, BOILING TEMPERATURE DOWN

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PASCAL → confined-fluid pressure transmission	BERNOULLI → pressure-flow relation under assumptions	ALTITUDE → pressure down, boiling temperature down
ARCHIMEDES → displaced-fluid weight and upthrust	PRESSURE COOKER → pressure up, boiling temperature up	PASCAL → confined-fluid pressure transmission

SYSTEM / DEVICE / INSTITUTION
- PASCAL → confined-fluid pressure transmission
- ARCHIMEDES → displaced-fluid weight and upthrust

DECISIVE TECHNICAL DISTINCTION
- BERNOULLI → pressure-flow relation under assumptions
- PRESSURE COOKER → pressure up, boiling temperature up

STATUS / UNIT / EVIDENCE LIMIT
- ALTITUDE → pressure down, boiling temperature down
- PASCAL → confined-fluid pressure transmission

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRESSURE COOKER → pressure up, boiling temperature up.

04

STAGE 04 THERMAL STATE AND TRANSFER LADDER

THERMAL STATE AND TRANSFER LADDER · THERMAL STATE · TRANSFER DUE TO TEMPERATURE DIFFERENCE · SPECIFIC HEAT · TEMPERATURE-CHANGE REQUIREMENT · LATENT HEAT · PHASE CHANGE WITHOUT TEMPERATURE CHANGE · DISTINCT ROUTES

TEMPERATURE → thermal state

HEAT → transfer due to temperature difference → **SPECIFIC HEAT → temperature-change requirement** → **LATENT HEAT → phase change without temperature change**

STARTING CONDITION / INPUT
- TEMPERATURE → thermal state
- HEAT → transfer due to temperature difference

SCIENTIFIC OR ENGINEERING MECHANISM
- SPECIFIC HEAT → temperature-change requirement
- LATENT HEAT → phase change without temperature change

OUTPUT / FAILURE BOUNDARY
- RADIATION → distinct routes
- TEMPERATURE → thermal state

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONDUCTION | CONVECTION | RADIATION → distinct routes.

05

STAGE 05 WAVE SOUND AND SPECTRUM BANDS

WAVE SOUND AND SPECTRUM BANDS · FREQUENCY X WAVELENGTH · MEDIUM REQUIRED · FREQUENCY CLASSES · DIFFERENT ELECTROMAGNETIC BANDS

WAVE → speed = frequency x wavelength → **SOUND → medium required** → **ULTRASOUND → frequency classes** → **RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA** → **RADAR / VLC → different electromagnetic bands**

CONCEPT / ARCHITECTURE
- WAVE → speed = frequency x wavelength
- SOUND → medium required

MECHANISM / APPLICATION
- ULTRASOUND → frequency classes
- RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA

READINESS / STATUS LIMIT
- RADAR / VLC → different electromagnetic bands
- WAVE → speed = frequency x wavelength

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA.

06

STAGE 06 MIRROR LENS AND FIBRE RAY LOGIC

MIRROR LENS AND FIBRE RAY LOGIC · CONVEX LENS · CONCAVE LENS · DENSER TO RARER + CRITICAL ANGLE · TOTAL INTERNAL REFLECTION

MIRROR → reflection

LENS → refraction → **CONVEX LENS → converging** → **CONCAVE LENS → diverging**

CONCEPT / ARCHITECTURE · **MECHANISM / APPLICATION** · **READINESS / STATUS LIMIT**

CONCEPT / ARCHITECTURE
- WAVE → speed = frequency x wavelength
- SOUND → medium required

MECHANISM / APPLICATION
- ULTRASOUND → frequency classes
- RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA

READINESS / STATUS LIMIT
- RADAR / VLC → different electromagnetic bands
- WAVE → speed = frequency x wavelength

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA.

06

STAGE 06 MIRROR LENS AND FIBRE RAY LOGIC

MIRROR LENS AND FIBRE RAY LOGIC | CONVEX LENS | CONCAVE LENS | DENSER TO RARER + CRITICAL ANGLE | TOTAL INTERNAL REFLECTION

MIRROR → reflection

LENS → refraction → CONVEX LENS → converging → CONCAVE LENS → diverging

CONCEPT / ARCHITECTURE
- MIRROR → reflection
- LENS → refraction

MECHANISM / APPLICATION
- CONVEX LENS → converging
- CONCAVE LENS → diverging

READINESS / STATUS LIMIT
- DENSER TO RARER + CRITICAL ANGLE → total internal reflection
- MIRROR → reflection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DENSER TO RARER + CRITICAL ANGLE → total internal reflection.

07

STAGE 07 CIRCUIT TOPOLOGY AND SAFETY BOARD

CIRCUIT TOPOLOGY AND SAFETY BOARD | WORK PER CHARGE | CHARGE PER TIME | OPPOSITION TO CURRENT | COMMON CURRENT | COMMON VOLTAGE | SHOCK PROTECTION

CONCEPT / ARCHITECTURE
- VOLTAGE → work per charge
- CURRENT → charge per time
- RESISTANCE → opposition to current

MECHANISM / APPLICATION
- SERIES → common current
- PARALLEL → common voltage
- FUSE / MCB → overload

READINESS / STATUS LIMIT
- EARTHING → shock protection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FUSE / MCB → overload | EARTHING → shock protection.

08

STAGE 08 ELECTROMAGNETIC CONVERSION LOOP

ELECTROMAGNETIC CONVERSION LOOP | CHANGING FLUX | INDUCED EMF | MECHANICAL TO ELECTRICAL | AC VOLTAGE CONVERSION | ELECTRICAL TO MECHANICAL | STEADY DC | NO ORDINARY TRANSFORMER ACTION

CONCEPT / ARCHITECTURE
- CHANGING FLUX → induced emf
- GENERATOR → mechanical to electrical

MECHANISM / APPLICATION
- TRANSFORMER → AC voltage conversion
- MOTOR → electrical to mechanical

READINESS / STATUS LIMIT
- STEADY DC → no ordinary transformer action
- CHANGING FLUX → induced emf

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MOTOR → electrical to mechanical; STEADY DC → no ordinary transformer action.

09

STAGE 09 QUANTUM AND NUCLEAR SPLIT PANEL

QUANTUM AND NUCLEAR SPLIT PANEL | THRESHOLD FREQUENCY | INTEGER-SPIN BOSONS MAY SHARE STATE | DIFFERENT MATTER/RADIATION CLASSES | NUCLEAR DECAY CLOCK

PHOTOELECTRIC → threshold frequency → BOSE-EINSTEIN → integer-spin bosons may share state → GAMMA → different matter/radiation classes → HALF-LIFE → nuclear decay clock → FISSION != FUSION

CONCEPT / ARCHITECTURE
- PHOTOELECTRIC → threshold frequency
- BOSE-EINSTEIN → integer-spin bosons may share state

MECHANISM / APPLICATION
- GAMMA → different matter/radiation classes
- HALF-LIFE → nuclear decay clock

READINESS / STATUS LIMIT
- FISSION != FUSION
- PHOTOELECTRIC → threshold frequency

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALPHA | BETA | GAMMA → different matter/radiation classes.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MOTOR → electrical to mechanical; STEADY DC → no ordinary transformer action.

09

STAGE 09 QUANTUM AND NUCLEAR SPLIT PANEL

QUANTUM AND NUCLEAR SPLIT PANEL | THRESHOLD FREQUENCY | INTEGER-SPIN BOSONS MAY SHARE STATE | DIFFERENT MATTER/RADIATION CLASSES | NUCLEAR DECAY CLOCK

PHOTOELECTRIC → threshold frequency → BOSE-EINSTEIN → integer-spin bosons may share state → GAMMA → different matter/radiation classes → HALF-LIFE → nuclear decay clock → FISSION != FUSION

CONCEPT / ARCHITECTURE
- PHOTOELECTRIC → threshold frequency
- BOSE-EINSTEIN → integer-spin bosons may share state

MECHANISM / APPLICATION
- GAMMA → different matter/radiation classes
- HALF-LIFE → nuclear decay clock

READINESS / STATUS LIMIT
- FISSION != FUSION
- PHOTOELECTRIC → threshold frequency

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALPHA | BETA | GAMMA → different matter/radiation classes.

10

STAGE 10 RELATIVITY OBSERVATION EVIDENCE CHAIN

RELATIVITY OBSERVATION EVIDENCE CHAIN | SPACETIME PREDICTION | GRAVITATIONAL WAVE | INSTRUMENT SIGNAL | SOURCE INFERENCE | DISTINCT ASTRONOMY LABELS | BOUNDED CLAIM, NOT OMNISCIENCE

RELATIVITY → spacetime prediction

GRAVITATIONAL WAVE → instrument signal → ANALYSIS → source inference → PULSAR → distinct astronomy labels

STARTING CONDITION / INPUT
- RELATIVITY → spacetime prediction
- GRAVITATIONAL WAVE → instrument signal

SCIENTIFIC OR ENGINEERING MECHANISM
- ANALYSIS → source inference
- PULSAR → distinct astronomy labels

OUTPUT / FAILURE BOUNDARY
- OBSERVATION → bounded claim, not omniscience
- RELATIVITY → spacetime prediction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CEPHEID | NEBULA | PULSAR → distinct astronomy labels.

11

STAGE 11 DEVICE TO VERIFIED-STATUS STAIRCASE

DEVICE TO VERIFIED-STATUS STAIRCASE | MECHANISM-SPECIFIC USE | ENABLING INSTRUMENT | NOT OPERATION | CALIBRATED OUTPUT | NOT AUTOMATICALLY VERIFIED PUBLIC OUTCOME

MATERIAL → DOPING → JUNCTION → DEVICE → VLC → mechanism-specific use → STANDARD / SCHEME → enabling instrument → TEST / SCHEDULE → not operation → CALIBRATED OUTPUT → not automatically verified public outcome

CONCEPT / ARCHITECTURE
- MATERIAL → DOPING → JUNCTION → DEVICE
- VLC → mechanism-specific use

MECHANISM / APPLICATION
- STANDARD / SCHEME → enabling instrument
- TEST / SCHEDULE → not operation

READINESS / STATUS LIMIT
- CALIBRATED OUTPUT → not automatically verified public outcome
- MATERIAL → DOPING → JUNCTION → DEVICE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CALIBRATED OUTPUT → not automatically verified public outcome.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | METROLOGY AND STANDARDS — ESSENTIAL WHEN PRECISION MEASUREMENT IS | ELECTRONICS AND DEVICE RESEARCH, LINKING SOLID-STATE PHYSICS TO USABLE | INDIA SEMICONDUCTOR MISSION | WHERE BASIC SEMICONDUCTOR PHYSICS BECOMES MANUFACTURING AND DESIGN POLICY. | DST NATIONAL QUANTUM MISSION | RELEVANT BECAUSE QUANTUM DEVICES, SENSING AND MATERIALS ALL REST | DAE/BARC ECOSYSTEM

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH
- CSIR-NPL: metrology and standards — essential when precision measurement is the real bottleneck.
- CSIR-CEERI: electronics and device research, linking solid-state physics to usable components.
- India Semiconductor Mission: where basic semiconductor physics becomes manufacturing and design policy.
- DST National Quantum Mission: relevant because quantum devices, sensing and materials all rest on advanced physics principles.

READINESS, UNIT OR STATUS LIMIT
- DAE/BARC ecosystem: policy-level continuation of nuclear and radiation-related physics in applied form.
- Power electronics, transformers and motors are central to India's electrification and industrial productivity.
- Laser-based tools matter in manufacturing, surveying, surgery and defence optics.
- Semiconductor devices determine telecom, computing, EV power systems, consumer electronics and strategic systems.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
- Radiation science supports diagnostics, therapy, sterilization and industrial measurement.
- CAUTION: Limitation 1: a device may work in the lab but fail at scale because thermal management, materials purity or measurement standards are weak.
- CAUTION: Limitation 2: public debate often jumps directly to final products while
- neglecting the physics-heavy middle layers such as metrology, materials and clean-room ecosystems.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels